

# Employee Management System

Database Design & Documentation

|                 |           |
|-----------------|-----------|
| Database Engine | MySQL     |
| Total Tables    | 12        |
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# 1. Project Overview

The Employee Management System (EMS) is a relational database designed to manage core HR operations for an organization. It covers employee records, departmental structure, salaries, attendance, leave management, payroll, bonuses, performance reviews, and project assignments. The schema follows a normalized 12-table design with foreign key relationships enforcing referential integrity across the system.

## Functional Areas Covered

- Core HR: departments, positions, employees
- Compensation: salaries, payroll, bonuses
- Attendance & Leave: attendance, leave\_types, leave\_requests
- Performance & Projects: performance\_reviews, projects, employee\_projects

# 2. Entity Relationships

| Child Table         | Foreign Key          | References                          |
|---------------------|----------------------|-------------------------------------|
| employees           | dept_id              | departments.dept_id                 |
| employees           | manager_id           | employees.emp_id (self-referencing) |
| employees           | position_id          | positions.position_id               |
| salaries            | emp_id               | employees.emp_id                    |
| attendance          | emp_id               | employees.emp_id                    |
| leave_requests      | emp_id               | employees.emp_id                    |
| leave_requests      | leave_type_id        | leave_types.leave_type_id           |
| leave_requests      | approved_by          | employees.emp_id                    |
| payroll             | emp_id               | employees.emp_id                    |
| bonuses             | emp_id               | employees.emp_id                    |
| performance_reviews | emp_id / reviewer_id | employees.emp_id                    |
| projects            | dept_id              | departments.dept_id                 |
| employee_projects   | emp_id / project_id  | employees / projects                |

### 3. Table Structures

#### departments

| Column    | Type          | Constraints        |
|-----------|---------------|--------------------|
| dept_id   | INT           | PK, Auto Increment |
| dept_name | VARCHAR(100)  | NOT NULL           |
| location  | VARCHAR(100)  |                    |
| budget    | DECIMAL(15,2) |                    |

#### positions

| Column      | Type          | Constraints        |
|-------------|---------------|--------------------|
| position_id | INT           | PK, Auto Increment |
| title       | VARCHAR(100)  | NOT NULL           |
| min_salary  | DECIMAL(10,2) |                    |
| max_salary  | DECIMAL(10,2) |                    |

#### employees

| Column                 | Type         | Constraints                    |
|------------------------|--------------|--------------------------------|
| emp_id                 | INT          | PK, Auto Increment             |
| first_name / last_name | VARCHAR(50)  | NOT NULL                       |
| email                  | VARCHAR(100) | UNIQUE, NOT NULL               |
| phone                  | VARCHAR(15)  |                                |
| hire_date              | DATE         | NOT NULL                       |
| dept_id                | INT          | FK -> departments              |
| manager_id             | INT          | FK -> employees (self)         |
| position_id            | INT          | FK -> positions                |
| job_title              | VARCHAR(100) |                                |
| status                 | ENUM         | Active / Inactive / Terminated |

#### salaries

| Column    | Type | Constraints        |
|-----------|------|--------------------|
| salary_id | INT  | PK, Auto Increment |
| emp_id    | INT  | FK -> employees    |

|                |               |             |
|----------------|---------------|-------------|
| amount         | DECIMAL(10,2) | NOT NULL    |
| effective_date | DATE          | NOT NULL    |
| currency       | VARCHAR(10)   | Default INR |

## attendance

| Column               | Type | Constraints                         |
|----------------------|------|-------------------------------------|
| attendance_id        | INT  | PK, Auto Increment                  |
| emp_id               | INT  | FK -> employees                     |
| date                 | DATE | NOT NULL                            |
| check_in / check_out | TIME |                                     |
| status               | ENUM | Present / Absent / Half Day / Leave |

## leave\_types

| Column            | Type        | Constraints        |
|-------------------|-------------|--------------------|
| leave_type_id     | INT         | PK, Auto Increment |
| leave_name        | VARCHAR(50) | NOT NULL           |
| max_days_per_year | INT         | Default 0          |

## leave\_requests

| Column                | Type | Constraints                   |
|-----------------------|------|-------------------------------|
| leave_id              | INT  | PK, Auto Increment            |
| emp_id                | INT  | FK -> employees               |
| leave_type_id         | INT  | FK -> leave_types             |
| start_date / end_date | DATE | NOT NULL                      |
| status                | ENUM | Pending / Approved / Rejected |
| approved_by           | INT  | FK -> employees               |

## payroll

| Column                           | Type          | Constraints        |
|----------------------------------|---------------|--------------------|
| payroll_id                       | INT           | PK, Auto Increment |
| emp_id                           | INT           | FK -> employees    |
| month / year                     | INT           | month 1-12         |
| basic_pay / deductions / net_pay | DECIMAL(10,2) |                    |

## bonuses

| Column       | Type          | Constraints        |
|--------------|---------------|--------------------|
| bonus_id     | INT           | PK, Auto Increment |
| emp_id       | INT           | FK -> employees    |
| amount       | DECIMAL(10,2) | NOT NULL           |
| reason       | VARCHAR(255)  |                    |
| date_awarded | DATE          | NOT NULL           |

## performance\_reviews

| Column      | Type         | Constraints        |
|-------------|--------------|--------------------|
| review_id   | INT          | PK, Auto Increment |
| emp_id      | INT          | FK -> employees    |
| reviewer_id | INT          | FK -> employees    |
| review_date | DATE         | NOT NULL           |
| rating      | DECIMAL(3,1) | CHECK 1-5          |
| comments    | TEXT         |                    |

## projects

| Column                | Type         | Constraints        |
|-----------------------|--------------|--------------------|
| project_id            | INT          | PK, Auto Increment |
| project_name          | VARCHAR(150) | NOT NULL           |
| start_date / end_date | DATE         |                    |
| dept_id               | INT          | FK -> departments  |

## employee\_projects

| Column              | Type         | Constraints       |
|---------------------|--------------|-------------------|
| emp_id / project_id | INT          | Composite PK, FKs |
| role_in_project     | VARCHAR(100) |                   |
| hours_allocated     | INT          | Default 0         |

## 4. Sample Data Summary

Each table has been populated with 10-12 sample rows to support testing and query practice. Employee records (emp\_id 101-110) form the central dataset that connects to salaries, attendance, leave requests, payroll, bonuses, performance reviews, and project assignments.

| Table               | Rows Inserted       |
|---------------------|---------------------|
| departments         | 10                  |
| positions           | 10 (after cleanup)  |
| employees           | 10 (emp_id 101-110) |
| salaries            | 10                  |
| attendance          | 10                  |
| leave_types         | 6                   |
| leave_requests      | 10                  |
| payroll             | 10                  |
| bonuses             | 10                  |
| performance_reviews | 10                  |
| projects            | 10                  |
| employee_projects   | 12                  |

## 5. Query Reference

The queries below demonstrate common search and reporting patterns used against the EMS schema, ranging from simple lookups to multi-table joins.

### Employee by name

```
SELECT * FROM employees WHERE first_name = 'Sanjeev';
```

### Employees by department

```
SELECT e.emp_id, e.first_name, d.dept_name
FROM employees e JOIN departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'Information Technology';
```

### Employee with manager (self-join)

```
SELECT e.first_name AS employee, m.first_name AS manager
FROM employees e
LEFT JOIN employees m ON e.manager_id = m.emp_id;
```

### Employees within a salary range

```
SELECT e.first_name, s.amount
FROM employees e JOIN salaries s
ON e.emp_id = s.emp_id
WHERE s.amount BETWEEN 500000 AND 1000000;
```

### Pending leave requests

```
SELECT e.first_name, lt.leave_name, lr.status
FROM leave_requests lr
JOIN employees e ON lr.emp_id = e.emp_id
JOIN leave_types lt ON lr.leave_type_id = lt.leave_type_id
WHERE lr.status = 'Pending';
```

### Full employee profile (5-table join)

```
SELECT e.first_name, d.dept_name, p.title,
s.amount, m.first_name AS manager
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
JOIN positions p ON e.position_id = p.position_id
JOIN salaries s ON e.emp_id = s.emp_id
LEFT JOIN employees m ON e.manager_id = m.emp_id;
```

The complete set of 15 search queries, along with the full schema and sample data, is available in the accompanying `ems_schema.sql` file.